

Nonnegative and positive factorizations

Full nonnegative rank of the quadratic correlation matrix

Difficulty: extreme **Importance:** broadly interesting

Status: Open

Rating rationale: Extreme because full nonnegative rank is much sharper than available exponential lower bounds for this family; broad importance concerns communication complexity and limitations of linear programming formulations.

Area: exact NMF and lower bounds for optimization formulations

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

For each integer $n \geq 3$, let C_n be the $2^n \times 2^n$ matrix indexed by $a, b \in \{0, 1\}^n$ and defined by

$$C_n(a, b) = (1 - a^\top b)^2.$$

Question

Is $\text{rank}_+(C_n) = 2^n$ for every $n \geq 3$? All entries, including those for $a^\top b > 1$, are fixed by this formula.

The matrix has ordinary rank $1 + n(n + 1)/2$ and is a submatrix of a slack matrix of the correlation polytope formed from valid, possibly redundant inequalities. The conjecture therefore asks for a large separation between ordinary and nonnegative matrix rank with consequences for linear programming representations.

References

Vandaele, Gillis, Glineur, and Tuytens, *Heuristics for Exact Nonnegative Matrix Factorization*, §6.4, Conjecture 4. Gillis, *Nonnegative Matrix Factorization*, §3.7, p. 96, using the name U_n for this fixed matrix.

Status check — 2026-09-10

Rechecked Baeckelant et al. v2, §6.7 and Appendix A.4 and Sergeev's July 2026 paper, and searched for a full-rank resolution. The former retains the prescribed-completion conjecture, including the unresolved $n=3$ case. Sergeev treats a partial unique-disjointness matrix whose remaining entries may vary; this does not determine the rank of the fixed matrix here. No full resolution was located.